Summary

- A measure which divides an array into four equal parts in known as **quartiles**.
- A measure which divides an array into ten equal parts is known as **deciles**.
- A measure which divides an array into hundred equal parts is known as **percentiles**
- $Q_2 = D_5 = P_{50}$ = Median
- Inter quartile range = $Q_3 - Q_1$
- QD = $\frac{Q_3 - Q_1}{2}$
- Harmonic mean = $\frac{n}{\frac{1}{X_1}+\frac{1}{X_2}+\frac{1}{X_3}+\cdots\frac{1}{X_n}} = \frac{n}{\sum\frac{1}{X}}$
- Mean Deviation for Individual series MD = $\frac{\sum|X-\bar{X}|}{n} = \frac{\sum|D|}{n}$
- The conditional probability of an event A given the occurrence of the event B is given by $P(A/B) = \frac{P(A \cap B)}{P(B)}$, $P(B) \neq 0$
- Baye's Theorem:
  If $E_1, E_2, E_3, ..., E_n$ are a set of n mutually exclusive and collectively exhaustive events with $P(E_i) \neq 0$ (i =1,2,3…,n), then for any arbitrary event A which is associated with sample space $S = \bigcup_{i=1}^{n} E_i$ such that $P(A) > 0$, we have

$$P(E_i / A) = \frac{P(E_i)P(A/E_i)}{\sum_{i=1}^{n} P(E_i)P(A/E_i)} \; ; \quad i = 1,2,3\ldots, n$$

where $P(A) = P(A) = \sum_{i=1}^{n} P(E_i)P(A/E_i)$

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